

Program Question_4

```
Implicit none
Real, Dimension(3)::u1,u2,u3
Real, Dimension(3)::v1,v2,v3
Real, Dimension(3)::e1,e2,e3
Real, Dimension(3)::proj,proj_1,proj_2
Real::dot12,dot13,dot23
Real::norm_v1,norm_v2,norm_v3
Integer::i
!Define the value
u1=[1.0,1.0,1.0]
u2=[1.0,1.0,0.0]
u3=[1.0,0.0,0.0]
!compute v1 and e1
v1=u1
norm_v1=Sqrt(Dot_Product(v1,v1))
e1=v1/norm_v1
!Compute v2 and e2
dot12=Dot_Product(u2,v1)
proj=(dot12/(norm_v1)**2)*v1
v2=u2-proj
norm_v2=Sqrt(Dot_product(v2,v2))
e2=v2/norm_v2

!Compute v3 and e3
dot13=Dot_Product(u3,v1)
proj_1=(dot13/(norm_v1)**2)*v1
dot23=dot_product(u3,v2)
proj_2=(dot23/(norm_v2)**2)*v2

v3=u3-proj_1-proj_2
norm_v3=Sqrt(Dot_product(v3,v3))
e3=v3/norm_v3

Write(*,*)"The orthonormal Basis e1,e2,e3 is_"
Write(*,*)"e1=",e1
Write(*,*)"e2=",e2
Write(*,*)"e3=",e3
```

End Program